CANSLIM, as O’Neil means it

Research and build spec for the market tab and the screener cards. How qp builds these values each night is in OPERATIONS.md; how the cards show them is in CARD-FIELDS.md and the Market tab in MARKET-TAB.md.

This document exists because most of these concepts have a plain-English name that means something *else* in common usage. Implementing the common reading produces numbers that look right, sit on a card, and are about a different thing. Every section below therefore leads with **what it is not**.

Sources are IBD’s and William O’Neil + Co’s own published descriptions, plus *How to Make Money in Stocks*. Where IBD’s published number has **changed over the years**, both are given and the choice is made explicit rather than hidden in a constant — see Thresholds that moved.

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1. The traps

Concept	What it actually is	What it is commonly mistaken for
Relative Strength (RS Rating)	A **percentile, 1–99**, of this stock's 12-month price performance against *every other stock*	**RSI** — a bounded oscillator comparing a stock to **its own** recent range. Says nothing about any other stock. Two names can both print RSI 70 while one leads the market and the other is a laggard bouncing in a downtrend
C in CAN SLIM	**Current quarter** EPS vs **the same quarter one year ago**	The latest quarter vs the *previous* quarter (sequential). Wrong: it reintroduces the seasonality the year-ago comparison exists to remove
A in CAN SLIM	**Annual** EPS growth, each year, over 3–5 years, plus ROE	A longer version of C. It is a different question: C is *acceleration now*, A is *durability*. A company can pass C and fail A (one good quarter) or pass A and fail C (a decelerating compounder)
Distribution day	*One session*: index closes **down ≥0.2%** on **volume higher than the prior session**	Any down day. Volume is the whole point — it is the footprint of institutions selling, not of the index drifting
Distribution day count	How many are **currently live** in a rolling **25-session** window, with expiry rules	A running total. A count that never expires only rises and is meaningless by March
Accumulation/Distribution Rating	IBD's **A–E letter** from 13 weeks of daily price/volume	The **Accumulation/Distribution *Index*** (Chaikin) — a completely different, unbounded cumulative line. Same words, different object
Follow-through day	A **confirmation**, on day **4+** of a rally attempt, with a **big gain on higher volume**	Any strong up day. The day count is what separates a real bottom from a dead-cat bounce
N — New high	Price emerging from a **base** into new high ground, on volume	Any 52-week high. A stock grinding to a high with no base and no volume is the opposite of the setup

Concept	What it actually is	What it is commonly mistaken for
Group rank	Rank of the stock's **industry group** (IBD uses **197**) by 6-month price performance	The GICS **sector** (11 of them). A sector is far too coarse — O'Neil's point is that leadership is a *group* phenomenon
M — Market direction	A **rule-based state machine** over the indexes	A sentiment read or a moving-average cross. O'Neil's whole claim is that it is mechanical
RS *Line*	The plotted ratio **stock ÷ index**, over time. Its *shape* is the signal — "RS line at a new high **before** price" is the MarketSmith tell	The **RS Rating**. Different objects: the Rating is today's percentile (a number 1–99), the Line is a curve. A stock can hold RS Rating 95 while its RS line rolls over — the Rating is backward-looking over 12 months, the Line turns first
Up/Down Volume Ratio	Up-day volume ÷ down-day volume over the last **50 days**. One number, ~1.0 neutral	The Acc/Dis Rating. Both measure institutional demand; U/D is a plain volume ratio over 50 days, Acc/Dis is a graded read of price *and* volume over 13 weeks. They disagree often and that disagreement is informative
Base stage	Which base this is **counted from the market bottom** — 1st, 2nd, late	Any consolidation. The count is the point: late-stage bases (3rd, 4th) fail far more often because the move is obvious by then
Group rank vs Group RS Rating	**One fact, two directions.** Rank 63 of 197 *is* the 68th percentile *is* the letter B+	Three separate measurements. They cannot disagree; building them as three fields invites a card that shows them doing so — see §9.2
Stock rank *within* its group	"**RS 1 of 13**" — the stock's RS Rating ranked against **only its own group's members**	The RS Rating itself. A stock can be RS 95 market-wide and 8 of 13 inside its group, i.e. the twelfth-best way to own the same theme. Nothing on our cards shows this today
N — the "new" thing	**Three** things: new product/service, new management, **or better industry conditions** — >95% of winners had one, *before* the advance	Two things (product + management), and a **news feed**. The third is always dropped, and it is the one we can compute without any news at all — see §10

2. M — the market model

This is the largest gap in the current system. `src/sideD/regime.js` today scores same-day and 1-week index moves into BULLISH/NEUTRAL/BEARISH. That is a short-term momentum blend. **It is not O'Neil's market model** and should not be presented as one. Both can exist; they must be named differently.

O'Neil's model is a **state machine over two indexes** (Nasdaq Composite and S&P 500), driven by two events.

2.1 The three states

IBD publishes these as *The Market Pulse*, with exactly three labels:

State	Meaning	Exposure implication O'Neil draws
Confirmed uptrend	A follow-through day has confirmed a rally and distribution has not yet accumulated	Buy breakouts
Uptrend under pressure	Still in an uptrend, but distribution days are stacking up	Stop initiating; tighten
Market in correction	The uptrend has broken	No new buying

Between a correction low and a follow-through day, the market is in a **rally attempt** — a fourth *internal* state, not one of the three published labels.

2.2 Distribution day — the exact rule

A session counts as a distribution day when, **on the same index**:

```
close ≤ prior_close × (1 − 0.002)        down 0.2% or more
AND volume > prior_volume                 on higher volume
AND the market is in a confirmed uptrend  (see below)
```

TWO FORMS, AND THEY DO NOT CONFLICT. The workshop states this plainly — *"closes LOWER in price than the prior session while trading volume increases"* — with no percentage at all. IBD's **0.2%** is the operational floor on that same rule, and it is there because without it roughly a quarter of all sessions qualify and the 5–6 cluster fires constantly. The floor is a named constant: **set it to 0 for O'Neil's plain form.**

THE WINDOW, likewise. The workshop gives the cluster as **5–6 days within 5–6 weeks**, which is 25–30 sessions. **25** is the tight end of his own range and IBD's published figure, so it is the default.

ONE INDEX IS ENOUGH. The workshop: *"You only need one major market index to hit this distribution threshold to shift the market status to a downtrend."* That is exactly why the published status here is the **worse** of the S&P 500 and the Nasdaq rather than a blend of the two.

The uptrend condition matters and is widely dropped. During a correction or a rally attempt there is no established uptrend to distribute *from*, so a heavy down day is not added to the count. A count that ignores this runs up during every correction and then reads as "extremely dangerous" on the day the bottom forms — precisely inverted.

Two removal rules, and a day leaves the count when **either** fires:

1. **Age** — it passes out of the trailing **25 sessions**.
2. **Price recovery** — the index **trades ≥5% above** that day's close.

SETTLED — 5%, and INTRADAY, not on a closing basis. The rule is that the index *trades* 5% above the distribution day's closing price, so it fires on the high, not the close. The first draft here said "closes ≥5% above", which is a stricter rule than IBD's and would hold days in the count that IBD had already dropped — the count would read more dangerous than the real one, on exactly the days a new uptrend is starting.

2.2b Stalling days — the form of distribution that is not a down day

This was missing from the first draft of this document and it is not a detail. IBD's distribution count includes a second shape:

```
STALLING (also called churning)
    volume is heavy — above the 25-day average, or above the prior day
    AND the index makes almost no upward progress (gain under ~0.2%)
    AND it closes in the lower half of the day's range
```

The logic is the same as a down day: institutions are selling into strength. The price does not fall because the selling is being absorbed — which is exactly what heavy volume with no progress *means*. A count that only looks for −0.2% closes misses this entirely, and it is the shape that shows up at tops, where the index grinds sideways on huge volume for a fortnight before it breaks.

A stalling day is a **distribution day** for counting purposes. It is flagged separately in the list so the two can be told apart, because they read differently on a chart.

2.3 The count → the state

Live distribution days	State
0–2	Confirmed uptrend
3–4	Confirmed uptrend, worth showing the count
5	**Uptrend under pressure**
6+	Typically precedes / accompanies **market in correction**

The count is not the only route into a correction: a decisive break of the rally low also ends the uptrend regardless of the count.

COUNTED PER INDEX, NOT POOLED. IBD tracks the Nasdaq Composite and the S&P 500 separately and publishes both counts. They diverge often — that divergence is information, because it says which half of the market is being sold. The status is driven by the **worse** of the two. Pooling them into one number would both double-count a day the two indexes shared and hide the divergence.

2.4 Rally attempt and follow-through day

CORRECTED FROM THE WORKSHOP (O'Neil, January 2010). The first version of this section anchored the count on "the first UP close after the low". That is a different anchor and therefore a different day number on every signal.

```
The ABSOLUTE MARKET BOTTOM = the LOWEST CLOSING PRICE of this correction
                             cycle. It never means zero — an index cannot
                             reach zero — it means the lowest local close
                             before the recovery begins.

Day 1 = the trading session IMMEDIATELY FOLLOWING that lowest close,
        whether that session is up or down.

Days 1, 2 and normally 3 are ignored ENTIRELY: early bounces off a bottom
fail often enough that the wait is the whole filter.

Follow-through day       = on day 4 or later of that rally attempt:
                             index gains ≥ 1.7%
                             AND volume > prior session's volume
```

SETTLED — 1.7%, which is O'Neil's own number, not a figure chosen here. He published 1% in the early editions, raised it to 1.5% in *How to Make Money in Stocks*, and to **1.7%** in *The Successful Investor* after program trading made 1% days ordinary. 1.7% is the last value he set, so it is the default.

The day window has a far end too, and the first draft here missed it: the follow-through is expected on **days 4–7**, occasionally as late as 10–11. Days 8–11 are flagged **late**, past 11 **very late**.

Flagged, not refused. An earlier version of this section said a rally attempt past day 11 "is not confirmed by it". Enforcing that creates a state the market cannot leave — an attempt that could never be confirmed sits in "correction" through an entire advance. The late ones are shown with their day number and their flag, and the reader decides.

The volume test is ONE comparison: above the prior session. An earlier draft here required volume above the index's own 50-day average as well. That was this document inventing a rule — it is a stricter variant some people apply, not the published one, which is *"in higher volume than the previous session"*.

It was caught by the first live run, and the cost is worth recording. With the extra test in place the model blocked the Nasdaq's August 2026 follow-through, the rally attempt ran to **day 23** unconfirmed, and the model published **"market in correction"** through a rally the S&P 500 had already confirmed with a clean day-4 follow-through. A rule stricter than the published one does not fail safe: it fails to the wrong answer, quietly, and in the direction of telling you not to buy.

The 50-day comparison is still measured and **reported** beside every follow-through, so the page can say how heavy the day was. It does not decide one.

- A follow-through **before day 4** is explicitly *not* one. Bounces in the first three days are the norm inside downtrends; the wait is the filter. **THE RESET IS THE SAME FACT, NOT A SECOND RULE.** A new lower **close** *is* a new bottom, so the anchor moves there and the count starts again at day 1.

And the anchor is a **close**, not an intraday low. A wick below the previous low that *closes above it* does not move the bottom and does not restart the count — the first version reset on the intraday low, which is a different bar and fired on days the market never actually closed through.

- A follow-through **fails** if the index subsequently undercuts the rally low. O'Neil is clear that not every FTD works — roughly a quarter fail — but that no major bottom has occurred *without* one. It is a **necessary, not sufficient** condition, and the tab should say so.
- The rally attempt **resets** if the low is undercut before an FTD appears.

2.5 RS — verified against the published formula

The implementation in `chart/relstrength.py` was checked against IBD's published description and **is correct**:

```
RS = 0.4·(P0/P63) + 0.2·(P0/P126) + 0.2·(P0/P189) + 0.2·(P0/P252)
```

equivalently `2·(P0/P63) + (P0/P126) + (P0/P189) + (P0/P252)`, all over 5 — 63/126/189/252 trading days being 3/6/9/12 months, with the most recent quarter carrying double weight. Percentile-ranked to 1–99.

Two things decide whether a reconstruction matches IBD's number, and both are choices this system has already made correctly:

- **The universe.** IBD ranks against all listed US stocks. Ranking against the Nasdaq only, or against the S&P 500, produces a different rating for the same stock. Ours uses Polygon's grouped-daily endpoint — every US ticker for the session — which is the right universe.
- **Adjusted prices.** A 2-for-1 split halves the raw close, and a 12-month performance measure on raw prices scores that stock at −50% and rates it 1. The strongest names in a bull market are exactly the ones that split.

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3. The card side — what to add

Current cards carry: score, price, change, gap %, rvol, ATR, regime, sector bias, sector, bias, catalyst, method. All are day-trading fields. Nothing on a card speaks to CAN SLIM except the membership tag.

Proposed additions, in the order a CAN SLIM reader looks at them:

Field	O’Neil’s letter	Source	Note
`rs`	**L**	**Already built** — `chart/relstrength.py`	Percentile 1–99. Reconstruction, not IBD’s number, and named so
`epsQtr`	**C**	SEC EDGAR XBRL	% change, current quarter vs **same quarter last year**
`epsQtrAccel`	**C**	EDGAR	Is this quarter’s growth **faster** than last quarter’s? O’Neil weights acceleration heavily
`epsAnnual`	**A**	EDGAR	3-year annual EPS CAGR
`roe`	**A**	EDGAR	Net income ÷ shareholders’ equity
`salesGrowth`	**S**/SMR	EDGAR	Latest quarter revenue vs year-ago quarter
`offHigh`	**N**	Price data (have)	% below the 52-week high — a Composite input in its own right
`groupRank`	**L/I**	Computed from our own universe	Rank of the stock’s industry group by 6-month performance. **Not** the 11-sector bias already on the card
`accDis`	**I**	Price/volume (have)	A–E from 13 weeks of price/volume. Must not be confused with the Chaikin index
`supply`	**S**	Free float × price	Small float = thinner supply = sharper moves

3.1 Free sources, named — and what each one cannot give

Need	Source	Free	Honest limit
Quarterly & annual EPS, revenue, net income, equity	**SEC EDGAR XBRL** `data.sec.gov/api/xbrl/companyfacts/CIK{...}.json`	Yes, official, **no key**	Needs a descriptive `User-Agent`. Filings land weeks after quarter-end, so a card in May may show a February filing — the as-of date must be on screen
CIK ↔ ticker	`www.sec.gov/files/company_tickers.json`	Yes	Refresh monthly
Number of funds owning, by quarter	**SEC Form 13F quarterly datasets**	Yes, official	Only managers over $100M file, and 45 days after quarter-end. So the count is real but lags a quarter — same lag MarketSmith has
Fund / bank / management ownership %	13F + Forms 3/4/5	Yes	Management % needs insider forms; more work than the fund count
Short interest	**FINRA** short interest files	Yes	Twice monthly, not daily
New CEO date	**SEC Form 8-K, Item 5.02**	Yes	Item extraction is text work, not a field lookup
Shares outstanding	EDGAR `dei:EntityCommonStockSharesOutstanding`	Yes	Exact
Float	—	**Partly**	No free feed publishes float directly. It is approximable as shares outstanding

Need	Source	Free	Honest limit
			minus insider and 5%-holder positions from Forms 4 and SC 13D/G. **This is the one number that will be an estimate, and it will be labelled as one**
Index OHLCV for the market model	Already have — Yahoo / Polygon	Yes	Nasdaq Composite, S&P 500
Whole-market percentiles (RS, group rank)	Already have — Polygon grouped daily	Yes	One call per session

No paid vendor is required for anything above. EDGAR is the authoritative source that the paid vendors resell, and 13F is how “number of funds” is knowable at all.

4. What goes where

Market tab

1. **The status**, as one of the three published labels, with the count that produced it and the rule that fired — never a bare word.
2. **Distribution day list** — the actual dates, per index, each with its % move and volume ratio, and when it expires. A count with no rows behind it cannot be checked.
3. **Rally attempt / follow-through** — which day of the attempt we are on, and, if an FTD fired, its date, gain and volume ratio.
4. **A stated caveat** that ~1 in 4 follow-throughs fail. The tab should not read as a signal to size up.
5. Keep the existing short-term momentum read, renamed so it is not mistaken for the O’Neil state.

Cards

A compact CAN SLIM strip: `RS 94 · C +38% · A +27% · Acc B · Grp 12/197 · −4% off high`, with anything unavailable shown as **unknown, never as zero**. A missing EPS is not a bad EPS, and a card that prints `C 0%` for a company that has not filed is worse than one that says nothing.

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5. Thresholds that moved

These are the numbers where honest sources disagree, because **IBD changed them**. Each is a named constant with both values recorded:

SETTLED: use O’Neil’s own numbers. Nothing here is chosen by us.

Rule	Value in use	Provenance
Follow-through gain	**1.7%**	O’Neil’s last published figure, *The Successful Investor*. He set 1% in the early editions, 1.5% in *How to Make Money in Stocks*, then 1.7% once program trading made 1% days ordinary. The last value he set is the one we use
Follow-through day window	**Days 4–7**, late to 10–11	O’Neil. Day 1 is the first up close off the low
Follow-through volume	**> prior session** — that is the whole test	IBD: *“in higher volume than the previous session”*. An earlier draft also demanded > the 50-day average; that was invented here, and the first live run showed what it cost — see §2.4
Follow-through day window	days 4–7, **flagged** late to 11 and beyond	Flagged, never refused: a rally attempt that can never be confirmed is a state the market cannot leave
Distribution recovery removal	**5%, intraday**	IBD: the index *trades* 5% above that day’s close. On the high, not the close
Distribution window	**25 sessions**	Consistent everywhere
Distribution day trigger	**−0.2%** on higher volume	Consistent everywhere
Stalling day	heavy volume, gain <0.2%, close in lower half	IBD, §2.2b
RS weighting	40/20/20/20 over 3/6/9/12 months	O’Neil’s published formula, §2.5
CANSLIM membership hold	**90 days**	Ours — this one is a system decision, not O’Neil’s, and is marked as such

The earlier draft of this section recommended 1.2% and a *closing* 5%. Both were this document choosing between sources rather than following O’Neil, and both are replaced above.

Each is still a **named constant, printed next to any signal that fires**. A follow-through day is a claim about the market; a claim whose definition is invisible cannot be checked against a chart afterwards. Configurable, but the default is O’Neil’s and changing it is a decision somebody has to make on purpose.

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6. What this deliberately does not do

- **It does not reproduce IBD's ratings.** EPS Rating, Composite and Accumulation/Distribution are commercial products with unpublished universes and weights. What is built here is a *reconstruction from the published description*, and it is named so — `rs`, not "IBD RS". A number that looked like IBD's and was not would be worse than no number.
- **It does not gate any screener.** Like the existing CANSLIM membership tag, these are labels. No tool's candidate list changes because of them.
- **It does not replace the day-trading fields.** CAN SLIM is a position-trading framework; the desk trades intraday setups. These fields are context, not entries.

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7. The card — exact spec (v2, from the MarketSmith panels)

Replaces the one-line version. The first draft compressed CAN SLIM into a single strip. That was wrong: C and A are *series*, and a single percentage throws away the shape — which is the part O'Neil reads. These are tables.

Two levels: a **row** in the list, and a **panel** on tap.

7.1 The row — what MarketSmith puts in its screen results

```
BGFV  Big 5 Sporting Corp                              Comp 96
      EPS 76 · RS 99 · Grp B+ · SMR B · A/D B+
      vol 1,737k  vs 50-day 1,044k  (+66%)
```

Six ratings, the same six MarketSmith's list view carries, plus volume against its own 50-day average — which is what MarketSmith means by `Volume +9%` and is the same quantity as the `rvol` already on our cards.

7.2 The panel — C, as a table

Eight quarters. `%Chg` is always against **the same quarter one year earlier**.

Qtr	EPS $	%Chg	Sales $M	%Chg	Margin
Jun-19	−0.03	n/a	241.0	0%	
Sep-19	0.30	+100%	266.2	0%	
Dec-19	0.05	+121%	244.1	−1%	
Mar-20	−0.22	n/a	217.7	−11%	−2.1%
Jun-20	0.39	+999%	227.9	−5%	+3.7%
Sep-20	1.31	+337%	305.0	+15%	+9.3%
Dec-20	0.83	+999%	290.6	+19%	+6.3%

Conventions taken from MarketSmith and kept:

- `n/a`**, not a number, when the year-ago quarter was a loss.** A percentage change from a negative base is arithmetic without meaning. MarketSmith prints `N/A`; so do we.
- **Capped at +999%.** Beyond that the number is noise; the point has been made.
- **Sales beside EPS, always.** O'Neil's warning is earnings growth without sales growth — margin games, buybacks, one-offs. The pair is the check.
- **After-tax margin**, because rising margin *and* rising sales is the combination he wants.

Derived and shown beside the table:

- **Accelerating?** Is each of the last 2–3 quarters' `%Chg` larger than the one before? O'Neil weights acceleration heavily and it is invisible in any single number.
- **How many of the last 8 quarters beat +25%.**

7.3 The panel — A, as a table

FY	EPS $	%Chg	Price high	Price low
2015	0.77		20	9
2016	0.78	+1%	15	8
…				
2021	1.74 est.	−25%		

Plus the three numbers MarketSmith puts beside it:

- **3-year EPS growth rate** (its `EPS Growth Rate`)
- **Earnings Stability** — a 0-99 measure of how much the earnings series *wobbles* around its trend. Low is good. O'Neil wants a straight line, not an average that happens to be high
- **Return on Equity** — his 17% floor

The price high/low column is there for a reason: it puts the earnings series next to what the stock did, which is the whole CAN SLIM claim.

7.4 The panel — I, with the history you asked for

MarketSmith's *Fund Ownership Summary* is a count of funds, by quarter:

Quarter	No. of funds
Mar-19	311
Jun-19	301
Sep-19	302
Dec-19	302
Mar-20	313
Jun-20	333
Sep-20	417
Dec-20	425

The trend is the signal, not the level. 311→425 over four quarters is institutions accumulating; a flat or falling count is the opposite, and the level alone tells you neither.

This is obtainable, free and official: SEC Form 13F. Every institutional manager over $100M files a 13F each quarter listing its US equity holdings, and the SEC publishes them as structured quarterly datasets. Counting the distinct filers holding a ticker in each quarter *is* this table. It is the same source the paid vendors use.

Beside it, from the same filings: **Funds %**, **Banks %**, **Management %**.

7.5 The panel — L, the group

From MarketSmith's Industry & Sector panel:

```
Industry group      Retail-Leisure Products      rank 63 of 197
Stocks in group     13
New highs / lows    4 / 0
This stock ranks    RS 1 of 13 · EPS 7 of 13 · A/D 4 of 13 · Comp 5 of 13
Top RS in group     BGFV 99 · HIBB 97 · HZO 96 · ONEW 96 · SPWH 82
```

Two separate facts, and both matter: **is the group strong**, and **is this stock the leader within it**. O'Neil buys the #1 or #2 name in a top group, not the cheapest name in it. The rank-within-group line is what makes that checkable, and nothing on our cards says it today.

7.6 The panel — S, supply

```
Shares outstanding  21.9 Mil
Float               21.3 Mil
Short interest      1.2 days, −21%
```

7.7 N — how, since you asked

N is four different things under one letter, and only some are computable:

N	Computable?	How
New **price high**	Yes	At/near a 52-week high — but the O'Neil meaning is *emerging from a base*, which needs base detection (phase 2)
New **management**	Yes, roughly	MarketSmith shows `New CEO 02/2021`. Source: SEC **Form 8-K Item 5.02**, which is filed for exactly this
New **product**	~~No~~ **Yes**	Superseded — openFDA, 8-K Items 8.01/7.01/1.01 and USASpending are all free and dated. See §10
New **industry conditions**	Partly	The group's rank *rising* is the measurable trace

~~Proposal: show % off 52-week high and New CEO date now; leave "new product" to the existing catalyst.~~ **Superseded by §10**, which makes all four computable through one scored pipeline. The "no" above was wrong: it assumed news had to come from a news feed. It comes from EDGAR.

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8. The market tab

The existing tab is kept. Its short-term index/sector read is useful and is not being replaced — it is renamed so it cannot be mistaken for the O'Neil state, and the O'Neil model is **added beside it** as its own block:

```
O'NEIL MARKET MODEL
  Status        Uptrend under pressure
  Because       5 distribution days live (>=5 is the threshold)
  Nasdaq        4 live      S&P 500   5 live
  Rally         n/a - in a confirmed uptrend since the FTD of 2026-07-18
  Rules in use  DD -0.2% on higher volume - 25 sessions - 5% intraday recovery
                FTD +1.7% on day 4-7, volume over the prior session

  DISTRIBUTION DAYS
  2026-08-27  S&P 500  -0.7%  vol x1.14   expires 2026-10-01
  2026-08-21  Nasdaq   -1.1%  vol x1.31   expires 2026-09-25
  ...
```

Every number carries the rule that produced it, and the distribution days are listed with their dates so the count can be checked against a chart. A status word with no rows behind it is not checkable, and this is a claim about when to stop buying.

A stated caveat sits under the follow-through line: roughly one in four follow-through days fails. It is a necessary condition for a bottom, not a sufficient one, and the tab must not read as permission to size up.

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9. Strength — market, sector, group, stock

You asked to check strength across the levels. There are **four**, not three, and each is computed by a *different* method. Treating them as one thing — “relative strength” — is the mistake this section exists to prevent.

O’Neil’s own numbers for why the middle two matter, from *How to Make Money in Stocks*: **37%** of a stock’s price move is attributed to its **industry group**, another **12%** to its **sector**. Roughly half the move is not about the company. That is the whole reason the group level exists on the card.

9.1 The four levels, and how each is computed

Level	The number	How it is computed	What it answers
Market	The O'Neil state (§2) + the index **RS line**	State machine over distribution days / rally attempt / FTD. The RS line is `index ÷ S&P 500` plotted	Should I be buying at all
Sector	Rank **1–33**	IBD's 33 sectors, ranked by aggregate price performance	Is the money in this half of the market — a coarse read
Group	Rank **1–197**	Least-squares curve fit on **summed member prices**, separate weightings per time period, ~**6-month** frame, cap-weighted	Is *this* the leading group. This is O'Neil's real level
Stock in group	"**RS 1 of 13**"	The stock's RS Rating ranked against only the members of its own group	Am I buying the leader or the laggard

The last row is the one nothing in our system shows today, and it is the one O'Neil is strictest about: buy the **#1 or #2** name in a top group, never the cheap laggard in it. A stock can carry RS Rating 95 — top 5% of the whole market — and still be **8 of 13** inside its own group, which means twelve better expressions of the same theme are on the screen next to it.

9.2 The precision point: three prints, one fact

From the MarketSmith panels, the same group appears as:

```
Industry 197 Rank    63          (rank, 1 = best, out of 197)
Group RS Rating      68          (percentile, 99 = best)
Group RS             B+          (letter band of that percentile)
```

These are **not three measurements**. `1 − 63/197 = 68%` — rank 63 of 197 *is* the 68th percentile, and B+ is that percentile in letter form. Building three separate fields would imply three independent reads and invite the card to show them disagreeing when they cannot.

Decision: compute the rank, derive the other two. Store one number.

9.3 Direction of the rank matters, and the two are opposite

- **Rank**: 1 is best, 197 is worst. Lower is stronger.
- **Rating / percentile**: 99 is best, 1 is worst. Higher is stronger.

Every place a group number is printed must carry its scale, because a bare "68" is excellent as a rating and mediocre as a rank. The cards will print `63 of 197` — never a bare rank — for exactly this reason.

9.4 The divisor is not permanent

IBD **restructured the group list from 197 to 145**. Any code that hardcodes 197, and any stored history that does not record which scheme it was computed under, silently rewrites its own past when the divisor changes. So:

- the divisor is **stored with the value**, not assumed;
- the card prints `63 of 197`, never `rank 63`;
- historical group ranks keep the divisor they were computed with, and a chart that spans the change says so rather than splicing two scales.

9.5 What we can actually build from free data

IBD's exact group definitions are proprietary — the 197-way split, and the "certain stocks within that industry" the curve fit runs on, are not published. What is reproducible:

Piece	Free?	How
Sector membership	Yes	Already in our data (`sector` on every card)
Group membership	**Approximated**	SIC code (EDGAR, free, on every filer), market-wide. The screeners' TradingView industry labels only fill symbols SIC cannot classify and never replace a SIC entry (F-21, 2026-10-02). Finer than sector, coarser than IBD's 197
Group strength rank	Yes	Rank our own groups by 6-month cap-weighted member performance — the same *method*, our own membership
Stock rank within group	Yes	Rank each member's RS Rating inside its group. **This is exact** — it needs only membership and our existing RS
Index RS line	Yes	`index ÷ S&P 500`, both already fetched

The honest limit, stated on the card: our group rank is *our* group rank, not IBD's, because the membership differs. The rank **within** the group and the number of members are exact given whatever membership we use, and those are the two facts O'Neil actually trades on. A card that showed "Group rank 63 of 197" while using a different group definition would be claiming IBD's number; it will print the divisor we actually used.

9.6 The index RS line

On a MarketSmith index chart the blue line on the Nasdaq Composite is the **Nasdaq's RS versus the S&P 500**, with the S&P 500 itself drawn behind it in black. Same construction as a stock's RS line, one level up: it says which index is leading, which is the earliest read on whether growth or defensives are being bought. It belongs on the **market tab**, not the cards.

---

10. N — the news algorithm (not a news tab)

You are right that N ends up being news, and right that a normal news tab is the wrong shape for it. This section is the algorithm.

10.1 What N actually is — and it is three things, not two

O'Neil studied the greatest winners 1953–1993 and found **more than 95%** had, before their big advance, one of:

1. a major **new product or service**
2. **new management**
3. an important **change for the better in the conditions of their industry**

The third is the one everybody drops, and we already compute it — a group climbing the rank in §9 *is* the measurable trace of changed industry conditions. It needs no news feed at all.

The fourth sense of "N", **new high**, is price and is already on the chart. It is the *confirmation*, not the cause.

10.2 Why the check period must be long — the core point

A news tab looks back **hours**. N must look back **24 months**.

The reason is O'Neil's own sequence: **the new thing comes first, the price move comes after.** A CEO who arrived nine months ago can still be the entire reason the stock is breaking out today. A tab that shows the last 24 hours can never see that, and will instead show an analyst note from this morning that means nothing.

So: **lookback 24 months, with decay, not a cutoff.**

Age of the event	Weight	Why
0–3 months	1.0	Fresh, and the move may be starting
3–6 months	0.9	O'Neil's sweet spot — old enough to be proven, young enough to still be running
6–12 months	0.7	Still the reason, but the obvious money is made
12–24 months	0.4	Fading into "the old story"
>24 months	0	Dropped

Note the shape: 3–6 months is **not** penalised versus today. A same-day headline has no price reaction yet, so it cannot be confirmed — it enters the card marked **unconfirmed**, not scored.

10.3 The pipeline

```
HARVEST  ->  CLASSIFY  ->  REACT  ->  PERSIST  ->  SCORE  ->  one line
(24 mo)      (3 buckets)   (price)   (RS line)    (0-100)    on the card
```

The step that makes this not a news tab is REACT. An event with no price-and-volume response *at the time it happened* is discarded. That is O'Neil's discipline applied to news: the story is only real if institutions acted on it. A new CEO announced into silence is not N.

10.4 HARVEST — the sources, all free

Bucket	Source	Free	What it gives
New management	**EDGAR 8-K Item 5.02**	Yes	*Structured.* The item code IS the classifier — mandatory, dated, no NLP needed. Highest precision source in the whole spec
New product	**openFDA** (drug approvals, device 510(k)/PMA)	Yes	A literal dated new product, for pharma/device names
New product	**EDGAR 8-K Item 8.01 / 7.01**	Yes	"Other events" / Reg FD — where product launches are announced
New contract	**EDGAR 8-K Item 1.01**	Yes	Material definitive agreement — a large new contract is a new revenue source
New contract	**USASpending.gov**	Yes	Government contract awards, dated, with the dollar amount
New industry conditions	**our own group rank** (§9)	Yes	Computed. Rank rising 3 months = the trace
Corroboration only	Finnhub / press-release RSS	Yes	Never the trigger — used to confirm and to supply the headline text

EDGAR is the spine because it is **structured, dated, mandatory and free**. News APIs are the corroboration layer, never the source of truth: a filing happened, a headline was written.

10.5 CLASSIFY — and what must be thrown away

Everything not in the three buckets is discarded. Named explicitly, because these are exactly the items that make a news tab feel busy and mean nothing:

Rejected	Why
Analyst upgrades / downgrades / price targets	Nothing is new **at the company**. This is opinion about the same facts
Index inclusion / rebalancing	Mechanical flow, not a business change
Earnings releases	That is **C and A**. Counting it in N double-counts the same fact
Splits, buybacks, dividends	Financial engineering, not a new product
Lawsuits, short-seller reports	News, but not the O'Neil sense of new
Secondary offerings / dilution	This is real and it belongs in **S** — supply going *up*. Routed there, not to N

10.6 REACT and PERSIST — the two price tests

REACT, measured on the event date + 2 sessions:

```
gap/move  in multiples of the stock's own 20-day ATR
volume    vs its own 50-day average
```

`R` is 0 when the move is inside one ATR on normal volume — and an event with `R = 0` is **dropped**, not shown small. This alone removes most of what a news tab would print.

PERSIST, measured today against the event date:

```
RS line higher than on the event day    P = 1.0
flat                                     P = 0.5
lower                                    P = 0.2
```

A catalyst the market has since given back is not the reason to buy today.

10.7 SCORE

```
N_event = W(class) x R(reaction) x D(age) x P(persistence)
N       = max over events, scaled 0-100
```

Class weights, in O'Neil's own order of importance:

Class	W
New product / service	1.00
New management (CEO / President)	0.90
New material contract	0.80
New industry conditions (group rank rising)	0.70
New high with no other N	0.40

Dedupe by `(class, date +/- 3 sessions)` — one event reaches us as an 8-K and as five headlines, and counting it six times would make the noisiest story the highest score.

10.8 How it presents on the card

One line, and a panel behind it:

```
N  86   New CEO - 2026-03-12 (5.8 mo) - +11.4% on 4.2x vol - RS line held
```

```
N - WHAT IS NEW                                        score 86
  2026-03-12  New management   8-K 5.02  +11.4%  4.2x vol  held    86
  2026-06-01  New product      FDA 510(k) +6.2%  2.1x vol  held    61
  2026-05..08 Industry cond.   group 141 -> 28   -         -       44
  ---
  unconfirmed (too new to score)
  2026-08-31  Contract         8-K 1.01  filed yesterday
  ---
  discarded: 34 items (no price reaction), 11 (wrong class)
```

The discarded count is printed on purpose. It is the evidence that the number is a filter and not a feed.

10.9 The difference, stated

A normal news tab	The N algorithm
Sorted by **time**	Sorted by **score**
Recent = important	A 9-month-old event can outrank this morning's
Every headline	Three buckets; everything else named and dropped
No link to price	No price-and-volume reaction, no entry
Window: hours	Window: **24 months**
You read it	It emits **one number and one dated line**
Duplicates across outlets	Deduped to the underlying event

---

11. Fitting it on the existing tools

Everything above is what to compute. This is where each piece lands in the system that already exists, and — as important — what it must not touch.

11.1 The rule that governs all of it

`src/sideA/canslim.js` already states the contract this system runs on, and it is the right one:

> It is a **LABEL, never a filter.** No tool's results change because of it. Reading it cannot alter which stocks a screener returns.

Everything in this document obeys that. Not one number here enters a score, a filter, or a rank. The reason is not tidiness: T1–T9 are nine independent experiments whose backtests are compared against each other. The moment an O'Neil number changes what a screener returns, every card captured before that change is measuring a different thing, and the comparison — the entire point of running nine tools — is gone.

So: **added to the card, never to the query.**

11.2 Compute once, read nine times

The market model is one fact about the market. Group ranks are one ranking of one universe. Computing either inside each of nine tools is nine chances to disagree, on nine different Polygon page loads.

The pattern already existed — `data/canslim-members.json`, written by T8, read by everyone, one-way, and a reader that cannot parse it carries on with no tags rather than failing a scan (that list was retired 2026-10-02; the letters below replace it). Three more files, same rules:

File	Written by	Contains	Refresh
`data/oneil-market.json`	**qp**	status, live distribution days per index with dates, rally-attempt state, last FTD, thresholds used	after each close, and intraday on demand
`data/oneil-groups.json`	**qp**	every group: rank, divisor, member count, member RS ranks	after each close
`data/oneil/<TICKER>.json`	nightly job	C table, A table, I history, S, N events	per field, see 11.5

Why qp is the writer. It already has what this needs and the nine tools do not: Polygon grouped-daily for the whole US universe, the parquet bar cache, and `chart/relstrength.py` — which §2.5 verified is already computing O'Neil's RS formula correctly, on the right universe, on adjusted prices. Rebuilding that in Node would be a second implementation of a formula we have already checked once.

Failure is silence. Every reader falls back to "no O'Neil data" and renders the card exactly as it renders today. The market model being stale must never be able to stop a scan.

11.3 The card — the row, and the panel behind it

Correction to this section's first draft. It reduced §7 to a badge reading `C92 A88 N86` and put "the tables" behind the existing Details fold. That contradicts §7, which exists *because* a single percentage throws away the shape O'Neil actually reads. A screener card column is 300–400px wide; an eight-quarter table with EPS, %Chg, Sales, %Chg and Margin does not fit in it and never will. Squeezing it in there is how the tables would have quietly become one number again.

So the split is the one §7 already stated — **a row, and a panel** — and the panel is a real full-width surface, not a fold inside a card.

On the card, always visible — the §7.1 row

Two lines, under the ticker line, above Volume & Float:

```
Comp 96    EPS 76 · RS 99 · Grp B+ · SMR B · A/D B+
           vol 1,737k vs 50-day 1,044k (+66%)  ·  grp 63/197 · RS 1 of 13
```

The same six ratings MarketSmith's list view carries, plus volume against its own 50-day average, plus the two group facts from §9. Tapping it opens the panel. This is what most mornings need and it is a block, not a badge.

The News section leads with N

```
News · N 86
▸ New CEO · 2026-03-12 (5.8 mo) · +11.4% on 4.2x vol · RS line held
---- recent headlines (3 weeks), unchanged ----
```

The existing headline list stays exactly as it is, three-week window and all. The scored 24-month answer sits **above** it. Two different objects in one place, which is where they belong: what is new about this company, and what was said about it this week. The full N table (§10.8) is in the panel.

The CANSLIM panel — where the tables actually live

Opened from the row. Full width, its own scroll, seven blocks in O’Neil’s own letter order so the panel reads as the method rather than as a data dump. The mechanism already exists on this page — `openChart()`, `toggleScoreBreakdown()` and `openTableInspector()` all open full surfaces from a card.

```
┌─ BGFV · Big 5 Sporting Corp ──────────────────── Comp 96 ─┐
│  EPS 76 · RS 99 · Grp B+ · SMR B · A/D B+                 │
├───────────────────────────────────────────────────────────┤
│  C — CURRENT QUARTERLY EARNINGS            §7.2           │
│  8 quarters. Qtr | EPS $ | %Chg | Sales $M | %Chg | Marg  │
│  %Chg always vs the SAME quarter a year earlier.          │
│  n/a on a loss base. Capped +999%.                        │
│  Accelerating: yes (3 of 3)   ·   Beat +25%: 5 of 8       │
├───────────────────────────────────────────────────────────┤
│  A — ANNUAL EARNINGS                       §7.3           │
│  FY | EPS $ | %Chg | Price high | Price low   (3-5 yrs)   │
│  3-yr EPS growth 34%  ·  Stability 12  ·  ROE 21%         │
│  ROE floor 17%: PASS                                      │
├───────────────────────────────────────────────────────────┤
│  N — WHAT IS NEW                           §10.8          │
│  date | class | source | reaction | persisted | score     │
│  unconfirmed (too new to score) listed separately         │
│  discarded: 34 no reaction, 11 wrong class                │
├───────────────────────────────────────────────────────────┤
│  S — SUPPLY                                §7.6           │
│  Shares out 21.9M · Float 21.3M · Short 1.2 days, -21%    │
│  U/D volume ratio 1.41 (50d)                              │
├───────────────────────────────────────────────────────────┤
│  L — LEADER OR LAGGARD                     §7.5, §9       │
│  Group Retail-Leisure Products     rank 63 of 197         │
│  Stocks in group 13   ·   New highs/lows 4 / 0            │
│  THIS STOCK   RS 1 of 13 · EPS 7 of 13 · A/D 4 of 13      │
│  Top RS in group  BGFV 99 · HIBB 97 · HZO 96 · ONEW 96    │
│  RS line: new high 2026-08-14, BEFORE price               │
├───────────────────────────────────────────────────────────┤
│  I — INSTITUTIONAL SPONSORSHIP             §7.4           │
│  Quarter | No. of funds     (8 quarters, from 13F)        │
│  311 → 425 over 4 quarters — the TREND is the signal      │
│  Funds 42% · Banks 9% · Management 14%                    │
├───────────────────────────────────────────────────────────┤
│  M — MARKET DIRECTION                      §2, §8         │
│  Uptrend under pressure · 5 distribution days live        │
│  (the one shared fact — same object the market tab shows) │
├───────────────────────────────────────────────────────────┤
│  Sources & dates: EDGAR 2026-08-30 · 13F Q2-26 ·          │
│  FINRA 2026-08-15 · groups 2026-08-31 close               │
└───────────────────────────────────────────────────────────┘
```

Why letter order and not importance order. The panel is the method, and the method is the mnemonic. Somebody checking a name works down the letters; a panel sorted by our idea of importance makes them hunt for A.

Every block carries its own as-of date, because they refresh on completely different clocks — 13F is quarterly and 45 days late by law, FINRA is twice a month, groups are daily. A panel with one timestamp at the top would be claiming a freshness five of the seven blocks do not have.

Market Context gains one line: the O'Neil status, beside the existing Regime/Long/Mid/Short rows, labelled so it cannot be read as the same claim.

Nothing is removed. Nothing that is on the card today moves.

11.4 The market tab — added beside, not over

`#tab-market` today is `#idx-grid` → `#market-analysis` (short/mid/long) → `#regime-card` (Final Regime) → the sector heatmap. Per your instruction, none of that is touched. Two additions:

- `#oneil-model`**, inserted above** `#market-analysis`**.** The block from §8: status, why, live counts per index, the rules in use, and the dated distribution-day table. It goes first because it is the one that decides whether to buy at all; the existing short/mid/long read is the finer texture underneath it.
- **A group table under the sector heatmap.** The heatmap is 15 sector ETFs — the coarse level. §9 is that leadership is a *group* phenomenon, so the top 20 and bottom 10 groups by our own rank go below it, each with its divisor and its member count. Same table styling, one level finer.

The existing "Final Regime" card keeps its logic and gets its title qualified so it cannot be mistaken for the O'Neil state — two market reads on one page that use the same words would be worse than only having one.

11.5 The nightly job, and why it cannot be in the request path

EDGAR is an HTTP fetch per company, rate-limited, and it must send a real User-Agent. openFDA and USASpending are the same shape. None of it can happen while a card is rendering.

So a **nightly job** walks the union of every tool's registry plus the CANSLIM list, and fills `data/oneil/<TICKER>.json`. Refresh rates follow how often the underlying fact can actually change:

Field	Source	Refresh
C, A	EDGAR XBRL company facts	weekly, and on a new 10-Q/10-K
I — fund count history	Form 13F	quarterly, after the 45-day filing deadline
S — float, shares out	EDGAR cover page	monthly
S — short interest	FINRA	twice monthly, on FINRA's own settlement calendar
N — events	8-K, openFDA, USASpending	nightly, 24-month window
RS, group rank, RS-in-group	qp / Polygon	after each close

A card that finds no file says **"not fetched yet"**, with the date it was last tried. It never blocks, and it never renders a blank that could be read as a zero.

11.6 The order to build it

1. **M — the market model in qp.** Distribution days including stalling, rally attempt, follow-through on O'Neil's settled numbers, three statuses, written to `data/oneil-market.json`. It is self-contained, it needs only index bars we already have, and it is the piece that changes behaviour on the most mornings.
2. **The market tab block.** Renders (1). Nothing else depends on it, and it is where the work becomes visible to you soonest.
3. **Groups and RS-in-group.** Built on `relstrength.py`, which already works. Gives the card its `grp 12/197` and `RS 1 of 13`, and the market tab its group table.
4. **The card badge line.** Renders (3) plus the RS already computed.
5. **EDGAR — C and A tables.** The largest piece, and the one whose tables you specified in §7.2 and §7.3.
6. **13F — the I history.**
7. **The N pipeline.** Last, because it is the only one that depends on several sources at once and on the price-reaction test.

11.7 The five things that must not happen

1. **No O'Neil number enters a score, filter or rank.** §11.1. This is the one that would quietly destroy the comparison between the nine tools.
2. **No network call in a card render.** Everything on a card comes from a file that a scheduled job wrote.
3. **Group ranks are not recomputed per tool.** One universe, one ranking, one writer.

4. **The existing three-week news window does not change.** The N pipeline is a separate object with a 24-month window. Widening the headline list to two years would turn the card’s News section into an archive.
5. **The market tab’s existing blocks are not rewritten.** Added beside, and the O’Neil state is labelled so the page never shows two market reads that use the same words for different claims.

---

12. The manifest — everything promised, and where each piece lands

You asked me to check everything I said, because §11 first shrank §7 into a badge. This section is the audit: **every element promised anywhere in this document**, its home, its source, and its build phase. If something is described above and is not in this table, it was dropped and that is a bug in the document.

Legend for **Home**: `ROW` = always visible on the screener card · `PANEL` = the CANSLIM panel (§11.3) · `NEWS` = the card’s existing News section · `CTX` = the card’s Market Context rows · `MKT` = the market tab.

12.1 M — market direction

#	Element	Home	Source	Rule	Phase
M1	Distribution day: close −0.2%+ on higher volume, **only in a confirmed uptrend**	MKT	Index bars	§2.2	1
M2	**Stalling day** — heavy volume, gain <0.2%, close in lower half	MKT	Index bars	§2.2b	1
M3	Live count per index, 25-session window	MKT	derived	§2.3	1
M4	Removal by age (25 sessions) **and by 5% intraday recovery**	MKT	derived	§2.2	1
M5	The three statuses: confirmed uptrend / under pressure / correction	MKT + CTX	derived	§2.1, §2.3	1
M6	Rally attempt — the internal 4th state, day 1 = first up close off the low	MKT	derived	§2.4	1
M7	**Follow-through day** — +1.7%, day 4+, volume **> the prior session** (the whole test). Days 8–11 flagged late, beyond that very late — flagged, never refused	MKT	derived	§2.4, §5	1
M6b	**The anchor** — the lowest **CLOSE** of the correction; day 1 is the session after it; a new lower close moves it and restarts the count	MKT	derived	§15.1	1
M8	FTD failure caveat — ~1 in 4 fail; necessary, not sufficient	MKT	text	§2.4	1
M9	**Dated distribution-day table** — every live day with date, index, %, vol ratio, expiry	MKT	derived	§8	1
M10	**Rules-in-use line** printed beside any signal that fires	MKT	constants	§5, §8	1
M11	Index **RS line** — index ÷ S&P 500	MKT	Index bars	§9.6	2
M12	Existing Final Regime kept, title qualified so two market reads never share words	MKT	existing	§11.4	2

12.2 The row — §7.1, always visible on the card

#	Element	Home	Source	Rule	Phase
R1	**Composite** rating	ROW	derived	§7.1, §6	4
R2	**EPS** rating	ROW	EDGAR	§7.1	5
R3	**RS** rating 1–99	ROW	`relstrength.py`	§2.5	3
R4	**Grp** RS — letter form of the group percentile	ROW	groups file	§9.2	3
R5	**SMR** — sales/margin/ROE	ROW	EDGAR	§7.1	5
R6	**A/D** rating A–E, 13 weeks price+volume	ROW	bars	§1, §7.1	4
R7	Volume vs its own 50-day average	ROW	bars	§7.1	3
R8	`grp 63 of 197` — **always with the divisor**	ROW	groups file	§9.3, §9.4	3
R9	`RS 1 of 13` — rank inside its own group	ROW	groups file	§9.1	3
R10	~~Existing `★ CANSLIM 47d` membership badge~~ — retired 2026-10-02 with T8's list	ROW	—	—	—

12.3 C — current quarterly earnings, §7.2

#	Element	Home	Source	Rule	Phase
C1	**8-quarter table**: Qtr, EPS $, %Chg, Sales $M, %Chg, Margin	PANEL	EDGAR XBRL	§7.2	5
C2	%Chg always against the **same quarter one year earlier**	PANEL	derived	§1, §7.2	5
C3	`n/a` when the year-ago quarter was a **loss** — never a percentage off a negative base	PANEL	derived	§7.2	5
C4	Capped at **+999%**	PANEL	derived	§7.2	5
C5	**Sales beside EPS, always** — earnings growth without sales growth is the warning	PANEL	EDGAR	§7.2	5
C6	After-tax **margin** column	PANEL	EDGAR	§7.2	5
C7	**Accelerating?** — each of the last 2–3 quarters' %Chg larger than the one before	PANEL	derived	§7.2	5
C8	How many of the last 8 quarters beat **+25%**	PANEL	derived	§7.2	5

12.4 A — annual earnings, §7.3

#	Element	Home	Source	Rule	Phase
A1	**FY table**: FY, EPS $, %Chg, Price high, Price low — 3–5 years	PANEL	EDGAR + bars	§7.3	5
A2	Price high/low column, so the earnings series sits next to what the stock did	PANEL	bars	§7.3	5
A3	**3-year EPS growth rate**	PANEL	derived	§7.3	5
A4	**Earnings Stability** 0–99 — the wobble around trend, low is good	PANEL	derived	§7.3	5
A5	**ROE**, against O’Neil’s **17% floor**, pass/fail shown	PANEL	EDGAR	§7.3	5
A6	Estimates marked `est.` and never mixed with reported	PANEL	EDGAR	§7.3	5

12.5 N — what is new, §10

#	Element	Home	Source	Rule	Phase
N1	**Scored N line** at the top of the News section	NEWS	pipeline	§10.8	7
N2	**Event table**: date, class, source, reaction, persisted, score	PANEL	pipeline	§10.8	7
N3	**24-month** lookback with decay 1.0 / 0.9 / 0.7 / 0.4	PANEL	—	§10.2	7
N4	**Unconfirmed** list — too new to have a price reaction	PANEL	pipeline	§10.2	7
N5	**Discarded counts** printed — the proof it is a filter, not a feed	PANEL	pipeline	§10.8	7
N6	New management — **8-K Item 5.02**	PANEL	EDGAR	§10.4	7
N7	New product — **openFDA** approvals, 510(k)/PMA	PANEL	openFDA	§10.4	7
N8	New product — 8-K Items **8.01 / 7.01**	PANEL	EDGAR	§10.4	7
N9	New contract — 8-K **1.01**, USASpending awards	PANEL	EDGAR, USASpending	§10.4	7
N10	New industry conditions — **our group rank rising**	PANEL	groups file	§10.1, §9	3
N11	**REACT** test — no move beyond 1 ATR on normal volume ⇒ dropped	PANEL	bars	§10.6	7
N12	**PERSIST** test — RS line vs the event day	PANEL	bars	§10.6	7
N13	Dedupe by (class, date ±3 sessions)	—	pipeline	§10.7	7
N14	Rejects named: analyst notes, index inclusion, earnings, splits/buybacks, suits	—	pipeline	§10.5	7
N15	**Secondary offerings routed to S**, not N — supply going up	PANEL(S)	EDGAR	§10.5	7
N16	**% off 52-week high**	ROW	bars	§7.7	3
N17	Existing three-week headline list — **unchanged, kept below N1**	NEWS	existing	§11.7	—

12.6 S — supply, §7.6

#	Element	Home	Source	Rule	Phase
S1	Shares outstanding	PANEL	EDGAR `dei:`	§3.1	5
S2	**Float — an estimate, and labelled as one** (no free feed publishes it)	PANEL	EDGAR 4 / 13D/G	§3.1	6
S3	Short interest — days to cover and % change	PANEL	FINRA	§3.1	6
S4	**U/D volume ratio** over 50 days — a different object from A/D	PANEL	bars	§1	4
S5	Existing Volume & Float block on the card — unchanged	ROW	existing	—	—

12.7 L — leader or laggard, §7.5 + §9

#	Element	Home	Source	Rule	Phase
L1	Group name and **rank of divisor**	PANEL + ROW	groups file	§7.5, §9.3	3
L2	Divisor **stored with the value**, because 197 → 145 happened	—	groups file	§9.4	3
L3	Stocks in group	PANEL	groups file	§7.5	3
L4	New highs / lows within the group	PANEL	bars	§7.5	4
L5	**This stock ranks**: RS, EPS, A/D, Comp — each `n of N`	PANEL	groups file	§7.5	3–5
L6	**Top RS names in the group**, listed	PANEL	groups file	§7.5	3
L7	**Sector rank 1 of 33** — the coarse level, kept distinct from group	PANEL	derived	§9.1	4
L8	**RS line at a new high before price** — the MarketSmith tell	PANEL	bars	§1, §7.5	4
L9	**Group table on the market tab** — top 20 / bottom 10 with divisors	MKT	groups file	§11.4	3
L10	Honest note: our membership is ours, not IBD’s	PANEL	text	§9.5	3

12.8 I — institutional sponsorship, §7.4

#	Element	Home	Source	Rule	Phase
I1	**Fund count by quarter, 8 quarters**	PANEL	SEC **13F**	§7.4	6
I2	The **trend** called out, not just the level	PANEL	derived	§7.4	6
I3	Funds % / Banks % / Management %	PANEL	13F + Forms 3/4/5	§3.1	6
I4	The 45-day statutory lag stated on screen	PANEL	text	§3.1	6

12.9 Cross-cutting

#	Element	Rule
X1	**Every block carries its own as-of date** — they refresh on different clocks	§11.3
X2	Panel is in **letter order** (C A N S L I M), because the panel is the method	§11.3
X3	Reconstructions are named as reconstructions — `rs`, never “IBD RS”	§6
X4	**Nothing enters a score, filter or rank.** Label, never filter	§11.1, §6
X5	**No network call in a card render** — every field comes from a written file	§11.7
X6	Computed once by qp, read by nine tools; failure = no data, never a failed scan	§11.2
X7	A missing file reads **“not fetched yet”** with the last attempt date — never a blank that looks like zero	§11.5
X8	Thresholds are **O’Neil’s**, named constants, printed next to any signal	§5

12.9b The market tab reflected per card — §14

#	Element	Home	Source	Rule	Phase
P1	The state, one compact line	CTX	`oneil-market.json`	§14.2	2
P2	**What this stock did on the live distribution days** — `up on 4 of 5, avg +0.9% vs index −0.8%`	CTX	market file + bars	§14.3	2
P3	The five dates behind a tap, so it is checkable against a chart	CTX	market file	§14.5	2
P4	Verdict: HOLDING UP / IN LINE / GIVING WAY	CTX	derived	§14.2	2
P5	**Sessions since the follow-through day**, banded early/established/late	CTX	market file	§14.3	2
P6	**Group rotation** — rank now vs 3 months ago, ▲into / ▼out of	CTX	groups file	§14.3	3
P7	The exposure implication in O’Neil’s words	CTX	text	§14.3	2
P8	Says so explicitly when there are **no** live distribution days	CTX	derived	§14.5	2

12.9c Everything explains itself — §13

#	Element	Rule
E1	Every label carries a tappable ⓘ — **tap, not hover**, because `title` does not exist on a touch screen	§13.1
E2	The definition card has **four parts, always**: what it means · how it is calculated · **what it is NOT** · source and as-of	§13.2
E3	`WHAT IT IS NOT` comes from the §1 trap table and is **mandatory**	§13.2
E4	A **Method page** in the tool: the trap table, every formula, the thresholds with provenance, the honest limits	§13.2
E5	No bare number — ranks print divisors, ratios print windows, percentiles print scales	§13.3
E6	No blank that reads as zero: `not fetched yet · last tried DATE`	§13.3, X7
E7	Every threshold that fires prints itself and whose number it is	§13.3, X8
E8	Existing `title` tooltips kept, but never the only copy of an explanation	§13.1

12.10 Phases — what each one delivers

Phase	Delivers	Depends on
1 ✅	M1–M10 — the whole market model, written to `oneil-market.json`	index bars only
2 ✅	The market tab O'Neil block (M5–M12) **and its per-card reflection P1–P5, P7, P8**	phase 1
3 ✅	Groups: R3–R4, R7–R9, L1–L3, L5(RS), L6, L9, L10, N10, N16, **P6**	`relstrength.py`
4 ✅	Ratings that need bars: **R6** (A/D, ours), **S4** (U/D ratio), **L8** (RS line before price), and the workshop's **divergence** test. R1/L4/L7 deferred — see below	phase 3
5 ✅	EDGAR: C1–C8, A1–A6, S1 — plus the WEEKLY base pattern	EDGAR fetcher
6	13F & FINRA: I1–I4, S2, S3	phase 5
7	The N pipeline: N1–N9, N11–N15	phases 3 and 5

Since this table was written, phase 6 has shipped I1, I2 and I4 (Form 13F, `chart/f13.py`, the nightly step "institutional sponsorship") and S3 (FINRA short interest, filled by the screener tools); S2 (the float estimate) and I3 (funds / banks / management %) are not built. Phase 7 is not built. What reaches the card: CARD-FIELDS #65–#83f.

E1–E8 are not a phase. Every phase ships its own definition cards with it; a field that arrives without its explanation is not finished. The Method page (E4) grows with each phase rather than being written once at the end.

Phases 1–4 need **no new data source at all** — index bars, Polygon grouped daily and `relstrength.py` are already in the system. That is where the market model, the whole group hierarchy and most of the row live, and it is why the build order in §11.6 starts there.

---

13. LOCKED — the card, and the rule that everything explains itself

The card spec (§7, §11.3, §12) is confirmed and locked. Changes to it from here are amendments with a reason recorded, not redesigns.

One requirement is added, and it applies to **every field in this document**:

> **Nothing is hidden, and every number explains its own meaning and its own calculation, inside the tool.**

Not in this file. This file is for building; the person using the tool at 07:40 is not going to open a repository.

13.1 The problem with how the card explains itself today

The card already carries good explanations — 14 of them in the card renderer alone, written as `title="..."` attributes:

```
title="volume against a normal day — 1.5× notable, 5× extreme"
title="shares actually available — under 20M is notable, under 10M is extreme"
```

A `title` attribute does not appear on a touch screen. There is no hover on Android Chrome, which is where this desk is actually read. So every one of those sentences is, in practice, invisible on the device it matters on — the work of explaining was already done and it is not reaching you.

That is not a CANSLIM problem, but CANSLIM triples the number of fields that need explaining, so it is fixed here rather than made worse.

13.2 The three levels of explanation

Every field gets all three. They are different jobs.

Level 1 — the label, tappable. Every label carries a small ⓘ. Tapping opens a definition card. Tapping, not hovering, so it works on the phone. The existing `title` text stays where it is — it costs nothing and it works on a desktop — but it is never the only copy of an explanation.

Level 2 — the definition card. Four parts, always the same four, because a definition with only the first part is what created every trap in §1:

```
┌ RS Rating ──────────────────────────────────┐
│                                             │
│ WHAT IT MEANS                               │
│ Where this stock's 12-month price           │
│ performance ranks against every other US    │
│ stock. 99 = top 1%. 50 = middle.            │
│                                             │
│ HOW IT IS CALCULATED                        │
│   RS = 0.4·(P0/P63) + 0.2·(P0/P126)         │
│      + 0.2·(P0/P189) + 0.2·(P0/P252)        │
│ 63/126/189/252 trading days = 3/6/9/12      │
│ months. The most recent quarter counts      │
│ double. Then ranked into a percentile       │
│ 1-99 against every US ticker.               │
│ Split-adjusted prices.                      │
│                                             │
│ WHAT IT IS NOT                              │
│ Not RSI. RSI compares a stock to its own    │
│ recent range and says nothing about any     │
│ other stock. Two stocks can both print      │
│ RSI 70 while one leads the market and the   │
│ other is a laggard bouncing in a downtrend. │
│                                             │
│ SOURCE · AS OF                              │
│ Polygon grouped daily · 2026-08-31 close    │
└─────────────────────────────────────────────┘
```

`WHAT IT IS NOT` **is mandatory and comes straight from §1.** That table exists because these concepts have plain-English names that mean something else. The trap column was written for the person reading the card, not for me.

Level 3 — the Method page. One page in the tool, reachable from the panel and from the market tab, holding every definition in one scrollable document: the §1 trap table, the formulas, the thresholds with O'Neil's provenance from §5, and the honest limits from §3.1 and §9.5. It is this document's content, rendered where it can actually be read.

13.3 "Everything visible" — what that rules out

- **No bare number anywhere.** A rank prints its divisor (`63 of 197`), a percentile prints its scale, a ratio prints its window (`U/D 1.41 · 50d`).
- **No blank that could be read as a zero.** Missing is `not fetched yet · last tried 2026-08-30`; not-applicable is `n/a` with the reason on tap.
- **No number without its as-of date** — per block, because they refresh on different clocks (§11.3, X1).

- **No reconstruction presented as the original.** `rs`, never “IBD RS”, and the definition card says so (§6, X3).
- **Every threshold that fired prints itself**, with whose number it is: `FTD +1.7% — O'Neil, The Successful Investor` (§5, X8).

---

14. The market tab, reflected in every card

This is the part that makes M worth having, and it is the one thing that cannot be solved by printing the market status on the card.

Why it matters, in O’Neil’s own terms: he found **three out of four stocks follow the general market direction**, and called M the most important letter — the one most investors ignore. His words: *“You can be right on every one of the factors in the last six chapters, but if you’re wrong about the direction of the general market, and that direction is down, three out of four of your stocks will plummet along with the market averages.”*

14.1 The trap: the same sentence on 150 cards is not information

The obvious build is to stamp `Market: uptrend under pressure` on every card. On a register day that is **150 identical lines**. A field with the same value on every row carries no information about any row — it is a page header that has been copied into the body 150 times, and after two mornings the eye stops seeing it.

So the market state appears on a card in **two parts**, and only the first is shared:

Part	Varies per card?	What it is
The state	No	One line, compact, the same everywhere. It is the header fact
This stock against it	**Yes**	What this specific stock did during the market events the tab is counting

The second is the point. It is computed from the market tab’s own output — the dated distribution days and the follow-through date — crossed with this stock’s bars.

14.2 The block, on every card

```
M — MARKET, AND THIS STOCK IN IT

  Market        Uptrend under pressure · 5 distribution days live
  Since FTD     32 sessions (follow-through 2026-07-18)

  ON THOSE 5 DISTRIBUTION DAYS
    This stock  up on 4 of 5   ·   avg +0.9%   vs index avg −0.8%
    Verdict     HOLDING UP — being accumulated while the index is
                being distributed

  GROUP ROTATION
    Retail-Leisure Products   rank 141 → 28 over 3 months   ▲ into

  WHAT O'NEIL DOES HERE
    Stop initiating new positions; tighten stops on open ones.
    A name holding up through distribution is a leader candidate
    for the next follow-through day.
```

14.3 The four per-card readings, and how each is computed

1. Behaviour on the distribution days — the strongest one.

The market tab already knows the exact dates of the live distribution days. For each, ask what *this* stock did on that same session:

```
held_up = count of live distribution days where
              stock's close-to-close return  >  index's return
avg_rel  = mean(stock return − index return) over those days
```

This is O'Neil's "leaders hold up during market pullbacks", made checkable. It is a **different number on every card**, computed from a market-level fact, and it is the single most useful thing the market model can tell you about one stock. Nothing in the system does this today.

2. Sessions since the follow-through day — the clock on the uptrend.

O'Neil buys early in a new uptrend. A breakout 5 sessions after an FTD and the same breakout 200 sessions after it are different propositions, and this is also the input to base-stage counting (§1) — bases are counted **from the market bottom**, so the FTD date is where that count starts.

```
sessions_since_ftd = trading days from the confirmed FTD to today
```

Banded on the card: `early (<25)` · `established (25–150)` · `late (>150)`.

3. Group rotation — the market's money, at this stock's level.

The market state is about the index; the group rank change is about where the money inside it is going. From §9's group file:

```
rank now  vs  rank 3 months ago      ▲ into / ▼ out of / flat
```

This is also N10 in the manifest — "new industry conditions" — and it is the same computation serving both letters. Computed once.

4. The exposure implication, in O'Neil's words, not a number.

Each of the three states carries the action O'Neil draws from it (§2.1). It is printed as text because it is a rule, not a measurement, and printed on the card because that is where the decision is being made.

14.4 Where each piece lives

Reading	Market tab	Card
The state and why	**Full** — status, per-index counts, dated distribution table, rules in use	One compact line
Distribution days	**The list**, with dates	**What this stock did on those dates**
Follow-through day	**The date, the day-count, the caveat**	Sessions since, banded
Groups	**Top 20 / bottom 10 table**	**This stock's group, and its rotation**
Exposure implication	Once, at the top	Once, per card

The tab holds the evidence; the card holds this stock's relationship to it. Neither repeats the other, and the card's version is a different number for every stock on the page.

14.5 The rules this must not break

- **Still a label, never a filter** (X4). "Held up on 4 of 5 distribution days" does not change the stock's score or whether it appears. It is the most tempting thing in this document to feed into a score, and feeding it in would make every card captured before today incomparable with every card captured after.
- **Computed from the shared file** (X6). The card does not recompute the market state; it reads `oneil-market.json` for the dates and does one comparison against bars it already has.
- **The dates are shown**, not just the count. "Up on 4 of 5" with the five dates behind a tap, so it can be checked against a chart.

- **It says when there are none.** In a confirmed uptrend with zero distribution days there is nothing to hold up through, and the block says exactly that rather than printing `0 of 0` or a blank.

---

15. From the workshop (O’Neil, January 2010) — what it changed

Three corrections to this document, all in the counting, plus one section that is still to build.

15.1 The anchor was wrong

	Was	Is
The bottom	the correction low	the **lowest CLOSE** of the cycle
Day 1	the first **up** close after it	the session **immediately after** the lowest close, up or down
The reset	the **intraday low** undercut	a **new lower close** — which *is* a new bottom

The reset is the part worth being careful about, because the two versions fire on different bars. A wick below the previous low that closes above it does not move the bottom; the old rule restarted a count the market had never broken. And the two are not separate rules at all: the anchor is “the lowest close so far”, so a lower close moves it and the count naturally starts again.

15.2 Two forms of the same rule, both recorded

Neither is a conflict, and both are now in the model with their provenance:

- **Distribution day** — the workshop says plainly *“closes lower on higher volume”*. IBD’s **−0.2%** is the operational floor on that. Configurable; zero gives the plain form.
- **The window** — the workshop says **5–6 weeks**; IBD says **25 sessions**. 25 is the tight end of his own range.

15.3 Still to build, from the workshop

The base pattern, which is what N and the buy point actually rest on:

Phase	What to detect
1 — Decline	the correction, typically in **three waves** down
2 — Base support	heavy volume with **no further price progress** — weeks where volume spikes or stays high while the price closes flat or tight. Institutional accumulation
3 — Handle	a minor controlled drift down, or tight sideways action, near the **upper** part of the base, before the pivot

And the divergence test, which is the sharpest version of what §14 already computes per card:

> While the index breaks to new lows, the stock **fails to make a new low** and reverses up. Its **RS line makes higher highs and enters new high ground BEFORE the price does.**

§14's "held up on 4 of 5 distribution days" is the same idea measured on distribution days. The full version — RS line at a new high before price — is `L8` in the manifest and belongs with base detection.

---

16. Phase 4, as built — and what was deferred, with the reason

16.1 Built

#	What	Where
S4	**U/D volume ratio** — up-day volume ÷ down-day volume, 50 sessions	card · Market Context
R6	**A/D** — a bounded, windowed 13-week read, banded A–E. **Ours**, with the raw number beside the letter	card
L8	**RS line at new high ground BEFORE price**	card
—	**Divergence** — the workshop's decoupling test	card

16.2 The two that carry the real weight

The RS line tell. `BEFORE` is the whole signal, and it is the thing an implementation loses first:

Case	Fires?	Why
RS line at a new high, price still in its base	**YES**	somebody is buying this while the market is not — legible *before* the breakout
RS line and price both at new highs	**no**	arithmetic. Price rose, the index did not, so the ratio rose. Says nothing the price chart did not
RS line off its high	no	the line is not leading anything

The audit tests the second case as hard as the first, because a version that could not tell them apart would still look right on the first.

Divergence needs the panic. All three must hold:

1. the **index** made a new low in the window
2. the **stock** did not
3. the stock is above where it sat on the index's worst close

Without (1) this measures nothing — a stock not making a new low in a calm month is not diverging from anything, it is simply a stock. So *"the index has not made a new low — nothing to diverge from"* is shown as a **different answer** from *"it followed the index down"*. Closes throughout, for the same reason the follow-through anchor is a close.

16.3 The A/D letter, and the trap it carries

`A/D` here is **not** the Accumulation/Distribution **Index** (Chaikin) — same words, different object, and it is in the §1 trap table for that reason.

	Chaikin's index	This
Window	all history, **cumulative**	fixed **13 weeks**
Bounds	unbounded	**−1 … +1**
Comparable between stocks	no	yes — normalised by the volume in the window

The letter is **ours**: IBD percentile-ranks its rating against the whole market and does not publish the boundaries, so this bands the raw score directly and prints the raw number beside the letter. Both facts travel with the number into the definition card, not just into this file.

16.4 Deferred, and why

#	What	Why not yet
R1	Composite rating	It blends EPS, RS, SMR, A/D and group. **EPS and SMR are phase 5** — a composite built now would be a partial one wearing a complete name, which is the exact failure §6 exists to prevent
L4	New highs / lows within a group	Needs 52-week highs for every group member, which is a universe-wide pass; it belongs with the group build, not the per-card one
L7	Sector rank 1 of 33	The machinery is the group ranking with a different key. Cheap, but it is the **coarse** level and §9 is explicit that the group is the one that decides — so it waits rather than competing for the same row

---

17. Phase 5 — the C and A tables, and the weekly base

17.1 The five EDGAR traps, each of which produces a NUMBER

Every one of these is wrong in a way that renders as a perfectly normal table:

#	Trap	What it does if missed
1	**C is year-over-year, never sequential**	a retailer's December against its September is a fact about Christmas
2	**Q4 is usually not filed as a quarter**	a 10-K reports the year, so Q4 exists only as FY − Q1 − Q2 − Q3. Reading only what is tagged quarterly **drops a quarter of every table**
3	**The same period is filed more than once**	restatements, amendments and comparative columns. The most recently **filed** value wins; taking the first match returns whatever was earliest in the array
4	**A percentage from a negative base is not a number**	this is how a screen surfaces a company that lost money last year as a 500% grower
5	**Revenue lives under four tags**	a filer who switched tags in 2021 has a hole in the middle of its table

Q4 is derived **only when all three earlier quarters are present**. A subtraction with a hole in it is a *wrong* number rather than a missing one.

17.2 The weekly base — and weekly is not a display choice

O'Neil taught bases on weekly charts and MarketSmith draws them weekly. It changes the answer:

- a base is **7–65 weeks**. On daily bars that is 35–325, and every shape test drowns in intraday noise
- *"heavy volume without price progress"* is a **week** closing flat on heavy volume. One day doing that is a session; five weeks of it is institutions absorbing what the decline shook out
- the handle is a **minor controlled drift of 2–8 weeks**. Measured daily it is a fortnight of wiggles with no shape

The three phases, each reported with its number rather than collapsed into a verdict — because O'Neil's own point about bases is that you *look* at them:

```
28-week base, 25% deep, 3 waves down, 4 accumulation weeks, handle 4w
Lip 100.00 (2024-03-01) → low 75.00 (2024-06-14)
Accumulation: 2024-05-10 2.1× · 2024-05-17 1.9× · …
Handle: 4 weeks, 7% deep, in the upper half
Pivot 104.60 · 3.2% away · 6 of 6 checks
```

A "handle" that gives back half the cup is the **cup failing**, not a handle — so `in_upper_half` is a named test and the panel says which test failed.

17.3 Where it lands

The **CANSLIM** ⤢ badge on every card opens a full-width panel in letter order — C, A, N, S, L, I, M. That is where §7's tables finally live: a card column is 300–400px and an eight-quarter table does not fit in one, which is how the tables quietly become a single percentage again.

I is not built and says so on the panel: fund ownership by quarter comes from Form 13F, which is phase 6. It is not estimated. (True when written; I has since been built from 13F: see §12.10.)

17.4 What the first live data check found

Run on the box, and it did its job:

Source	Verdict
yahoo, hybrid_yahoo, ^GSPC, ^IXIC	ok
polygon, alpaca, hybrid	DOWN — **no keys in the shell's environment**
polygon grouped-daily (RS universe)	DOWN — nothing cached
group ranks, industry map	not written yet

And it exposed a flaw in **the check itself**: qp then ran under systemd with `EnvironmentFile=~/trade-desk.env` (it now runs under pm2 and reads `quant-platform/.env`: OPERATIONS.md), a bare shell does not, so the runner was reporting a different environment from the one that actually runs. A red light over a green system is worse than no light — the next real failure is the one nobody believes. The runner now loads the same file and says which.

It also now reports when **fewer than two feeds can be compared**: with one working feed nothing is checking whether it is *right*, only that it answered, and that state was previously invisible because the agreement rows simply did not appear.

Every failing check now carries a **fix**, not just a diagnosis.